Assumptions:

Assumptions:

Assumptions class provide static methods.

assumeTrue:

assumeTrue(boolean condition)
assumeTrue(boolean condition, String message)

assumeFalse:

assumeFalse(boolean condition)
assumeFalse(boolean condition, String message)

assumingThat:

assumingThat(boolean condition, Executable executable)

- We can provide conditions, and if conditions is true then only Test case will proceed.
- If the condition fails, the test is skipped(aborted).

<u>assumeTrue:</u>

Skips the test if condition is false.

```java
package concepts.Assumptions;

import org.junit.jupiter.api.Assumptions;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import org.opentest4j.TestAbortedException;

public class AssumptionTest {

    @BeforeEach
    void setup(){
        System.setProperty("env", "Prod");
    }

    @Test
    void testForDevelop() {

Assumptions.assumeTrue("Dev".equals(System.getProperty("env")), "Test case is not eligible for this env");
    }

    @Test
    void testForProduction() {

Assumptions.assumeTrue("Prod".equals(System.getProperty("env")));
        System.out.println("Test case can proceed");
```

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```

    }
}
```

When Jupiter engines runs a particular test, and when this ("assumeTrue") method is encountered, internally it simply check if assumption is false then it throws an exception.

```
public static void assumeTrue(boolean assumption, String message) throws TestAbortedException {
    if (!assumption) {
        throwAssumptionFailed(message);
    }

}
```

Output:

```
AssumptionTest (concepts.Assumptions)                                  10 ms
    testForProduction()                                                 8 ms
    testForDevelop()                                                    2 ms

Tests passed: 1, ignored: 1 of 2 tests – 10 ms

org.opentest4j.TestAbortedException: Assumption failed: Test case is not eligible for this env

    <2 internal lines>
    at concepts.Assumptions.AssumptionTest.testForDevelop(AssumptionTest.java:17) <1 internal line>
    at java.base/java.util.ArrayList.forEach(ArrayList.java:1596)
    at java.base/java.util.ArrayList.forEach(ArrayList.java:1596)
```

assumeFalse:

Runs the test only if condition is false.

```

@Test
    void testMethod() {
        Assumptions.assumeFalse( Runtime.version().feature()==17,
                "Test case is not eligible for this java version");
    }



```

Output:

```
Tests ignored: 1 of 1 test – 9 ms

/Users/shrayanshjain/Library/Java/JavaVirtualMachines/ms-21.0.8/Contents/Home/bin/java ...

org.opentest4j.TestAbortedException: Assumption failed: Test case is not eligible for this java version

    <2 internal lines>
```

assumingThat:

- Execute the code block only if the condition is True.
- If False, rest of the testcase still runs.

```
@Test
    void testAssumingThat(){
        Assumptions.assumingThat(
                Runtime.version().feature()==21,
                () -> {
                    System.out.println("This code block runs only if condition is true");
                }
        );
        System.out.println("This code will runs always");
    }
```

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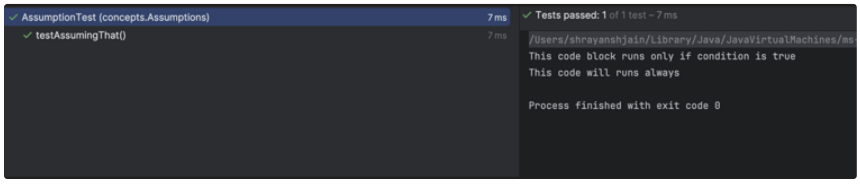
```

Output:



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